

TØ 2 – Mikroøkonomi 2 – 2024

Problem 1

Penalty kicks. Consider a penalty kick situation in soccer. A well-kicked ball takes a split second to reach the goal. Therefore, the goalie has to decide before the ball is kicked which side to jump to. The kicker must decide in his run up which way to kick the ball, before he sees to which side the goalie is leaning. Suppose the kicker is right-footed, which makes it easier to kick to the right side of the goal. The following table gives the payoffs¹.

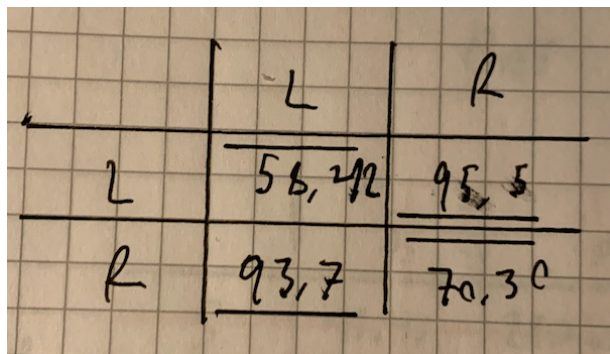
Kicker (K)\Goalie (G)	L	R
L	58,42	95,5
R	93,7	70,30

Find the Nash equilibrium of this game.

Pure strategy

There is no pure Nash equilibrium as shown down below

Why? Because one player would deviate



	L	R
L	58,42	95,5
R	93,7	70,30

Mixed strategy

		q		$(1-q)$
		L	R	
p	L	58, 42	95, 5	
$(1-p)$	R	93, 7	70, 30	

$$58q + 95(1 - q) = 93q + 70(1 - q) \Leftrightarrow q^* = \frac{5}{12} \approx 0.42$$

And

$$42p + 7(1 - p) = 5p + 30(1 - p) \Leftrightarrow p^* = \frac{23}{60} \approx 0.38$$

Therefore, the equilibria are

$$NE = \left\{ \left(\frac{23}{60}, \frac{37}{60} \right), \left(\frac{5}{12}, \frac{7}{12} \right) \right\}$$

Meaning that the kicker plays a mixed strategy of $\left(\frac{23}{60}, \frac{37}{60} \right)$

And the goalie plays a mixed strategy of $\left(\frac{5}{12}, \frac{7}{12} \right)$

Payoff is

$$u_k = 58pq + 95p(1 - q) + 93(1 - p)q + 70(1 - p)(1 - q) = 79.583$$

$$u_g = 42pq + 5p(1 - q) + 7(1 - p)q + 30(1 - p)(1 - q) = 20.417$$

Because $q = 0.42$ and $p = 0.38$

Problem 2

Consider again the problem of penalty kicks in soccer. Suppose now that the goalie greatly improves his skill at saving penalties struck at the right side, so that the success rate for the kicker is reduced by 20 percentage points to 50 percent. That is, the new game looks as follows.

Kicker (K)\Goalie (G)	L	R
L	58,42	95,5
R	93,7	50,50

- (a) Determine the NE.
- (b) How do the equilibrium strategies and the expected payoffs for the kicker and the goalie compare with the old equilibrium from the first problem in this problem set? Is this in line with your intuition?

a)

$$58q + 95(1 - q) = 93q + 50(1 - q) \Leftrightarrow q = \frac{9}{16} \approx 0.56$$

$$42p + 7(1 - p) = 5p + 50(1 - p) \Leftrightarrow p = \frac{43}{80} \approx 0.54$$

MSNE is then

$$(p, (1 - p)); (q, (1 - q)) = \left(\left(\frac{43}{80} \right), \left(\frac{37}{80} \right); \left(\frac{9}{16} \right), \left(\frac{7}{16} \right) \right)$$

b)

So now, the goalkeeper has been training a lot in jumping to the right because the kicker is better shooting in that side.

However, he will use his left side more than before.

Why? Because the kicker responds to the fact that the keeper has trained right side by kicking more frequent to the left.

However, payoff is

$$u_k = 58pq + 95(1 - q)p + 93(1 - p)q + 50(1 - p)(1 - q) = 74.19$$

$$u_g = 42pq + 5(1 - q)p + 7(1 - p)q + 50(1 - p)(1 - q) = 25.81$$

Because $p = 0.54$ and $q = 0.56$

Thus, the goalies payoff has increased

Problem 3

Interaction between a railway passenger and the railway operator can be described as follows

1 \ 2	<i>don't inspect</i>	<i>inspect</i>
<i>pay</i>	1, 1	2, 0
<i>freeride</i>	2, 0	α , 1

(a) Determine the equilibrium probabilities $(p^*(\alpha), q^*(\alpha))$ with which

- the passenger chooses *pay* and
- the operator chooses *don't inspect*

for general $\alpha \leq 0$.

a)

This is the mixed Nash equilibrium. I solve it by

$$1q + 2(1 - q) = 2q + \alpha(1 - q) \Leftrightarrow q^* = \frac{\alpha - 2}{\alpha - 3} = \frac{2 - \alpha}{3 - \alpha}$$

And

$$1p + 0(1 - p) = 0p + 1(1 - p) \Leftrightarrow p^* = \frac{1}{2}$$

Thus, the Nash equilibria are

$$NE = \{(p, 1 - p), (q, 1 - q)\} = \left\{ \left(\frac{1}{2}, \frac{1}{2} \right), \left(\frac{2 - \alpha}{3 - \alpha}, 1 - \frac{2 - \alpha}{3 - \alpha} \right) \right\}$$

b)

(b) **Draw** the players' best response correspondences for $\alpha = 0$.

If $\alpha = 0$, q^* becomes

$$q^* = \frac{2 - \alpha}{3 - \alpha} = \frac{-2}{-3} = \frac{2}{3}$$

The equilibria are now

$$NE \left\{ \left(\frac{1}{2}, \frac{1}{2} \right), \left(\frac{2}{3}, \frac{1}{3} \right) \right\}$$

I now find the best response for player 1. He will prefer pay to pay when $v_1(P, q) > v_1(F, q)$, i.e.,

$$1q + 2(1 - q) > 2q + \alpha(1 - q) = 2q + 0(1 - q) = 2q \Leftrightarrow q^* < \frac{2}{3}$$

Implying that

$$BR_1(q) = \begin{cases} p = 1, & q < \frac{2}{3} \\ p \in [0,1], & q = \frac{2}{3} \\ p = 0, & q > \frac{2}{3} \end{cases}$$

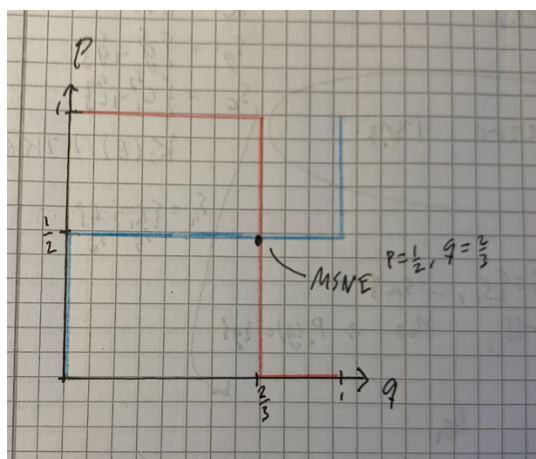
And player 2 will prefer DI to I when

$$1p + 0 \cdot (1 - p) > 0 \cdot p + 1 \cdot (1 - p) \Leftrightarrow p > \frac{1}{2}$$

Implying that

$$BR_2(p) = \begin{cases} q = 0, & p < \frac{1}{2} \\ q \in [0,1], & p = \frac{1}{2} \\ q = 1, & p > \frac{1}{2} \end{cases}$$

Graphically:



c)

- (c) Briefly discuss how above probabilities respond to changes in α (e.g., reflecting different penalties for a passenger caught freeriding).

Since $\alpha \leq 0$, lower α implies fewer inspections from the operator

For example, if $\alpha = -5$, the operator will play a mixed strategy of

$$1q + 2(1 - q) = 2q - 5(1 - q) \Leftrightarrow q = 0.875$$

So, operator will play a mixed strategy of

$$(0.87, 0.13)$$

And if $\alpha = -1$, operator will play a mixed strategy of

$$0 = 1q + 2(1 - q) = 2q - 1(1 - q) \Leftrightarrow q = 0.75$$

So,

$$(0.75, 0.25)$$

The passenger will play a mixed strategy of (0.5, 0.5) no matter the value of α meaning there is a 50/50 of him buying a ticket no matter the fine for being busted without a ticket Why?

Well, the operator adjusts his inspection according to alpha: more negative $\alpha \Rightarrow$ inspect less such that the passenger is indifferent between paying and freeriding

Problem 4

Two rival companies, X-Co and Y-Co, are vying for a valuable government contract in the country of Zlandia. The contract holds a value of 20 for each competitor, with the winning company receiving 20 and the losing company gaining nothing. In Zlandia's customs,

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the norm for a bribe involves presenting either 3 or 15 baskets of exotic fruits. (These numbers, 3 and 15, are regarded as fortuitous, while other figures are considered ill-fated and unacceptable.) The cost of each basket is 1.

Both companies have the option to offer a bribe discreetly, and the one offering the highest bribe will secure the contract. If both companies opt not to offer a bribe or if their bribe amounts match, each will have an equal chance of clinching the contract.

- (a) Find the unique pure-strategy Nash equilibrium for this 3x3-game.
- (b) Times have changed in Zlandia, so that 7 now also counts as a lucky number, meaning that 7 baskets of exotic fruits are also acceptable as a bribe. Write down the normal form game when the companies have 0 (no bribe), 3, 7 and 15 baskets as possible bribes. Check if there is one or more pure-strategy Nash equilibria.
- (c) Find a mixed-strategy Nash equilibrium of the game you set up in (b), where the equilibrium strategy assigns zero probability to bribing with 3 baskets.

a.

Found by the BR method and the general formula

$$P(win) \cdot 20 - c$$

So, (T, F) , for example:

Payoff player 1

$$0 \cdot 20 - 3 = -3$$

Payoff player 2

$$1 \cdot 20 - 15 = 5$$

A \ B	Z	T	F
Z	10, 10	0, 13	0, 5
T	17, 0	7, 7	-3, 5
F	5, 0	5, -3	-5, -5

b.

Clearly, no PSNE

How about mixed?

A \ B	Z	T	S	F
Z	10, 10	0, 13	0, 13	0, 5
T	17, 0	7, 7	-3, 3	-3, 5
S	13, 0	13, -3	3, 3	-7, 5
F	5, 0	5, -3	5, -7	-5, -5

Notice that the probability of playing T is 0 according to the assignment description and that the payoffs are symmetric, implying that we only need to figure out q_1 and q_2 as $q_1 = p_1, q_2 = p_2$ because of symmetry

Player 1 is indifferent between Z and S if and only if

$$10q_1 + 0q_2 + 0(1 - q_1 - q_2) = 13q_1 + 3q_2 + (-7)(1 - q_1 - q_2)$$

$$\Rightarrow 10q_1 = 13q_1 + 3q_2 + (-7)(1 - q_1 - q_2) \Rightarrow 7 = 10q_1 + 10q_2$$

And indifferent between Z and F if and only if

$$10q_1 + 0q_2 + 0(1 - q_1 - q_2) = 5q_1 + 5q_2 + (-5)(1 - q_1 - q_2) \Rightarrow 5 = 10q_2 \Leftrightarrow q_2 = \frac{1}{2}$$

And q_1 can now be calculated

$$7 = 10q_1 + 10q_2 = 10q_1 + 10 \cdot \frac{1}{2} \Leftrightarrow 7 = 10q_1 + 5 \Leftrightarrow 2 = 10q_1 \Leftrightarrow q_1 = \frac{1}{5}$$

MSNE is therefore

$$(q_1, 0, q_2, 1 - q_1 - q_2) = \left(\frac{1}{5}, 0, \frac{1}{2}, 1 - \frac{1}{5} - \frac{1}{2}\right) = \left(\frac{1}{5}, 0, \frac{1}{2}, \frac{3}{10}\right)$$

Problem 5

Cournot oligopoly model. There are 2 firms. They choose simultaneously quantities $q_1 \in \mathbb{R}_0^+$ (firm 1) and $q_2 \in \mathbb{R}_0^+$ (firm 2). Let $Q = q_1 + q_2$. The inverse demand function is $p(Q) = 1 - Q$. (That is, the aggregate demand at a given price p is $Q(p) = 1 - p$.) Each firm produces at a constant marginal cost $1 > c > 0$, i.e. the cost of producing q_i units is $c q_i$.

- (a) What is the best response of firm i to the competitor's quantity q_{-i} ?
- (b) What is the most firm i would ever be willing to produce?
- (c) What strategies survive one round of elimination of strictly dominated strategies (IESDS)?
- (d) What strategies survive two rounds of IESDS?
- (e) What strategies survive three rounds of IESDS?
- (f) Determine the best response function of each firm.

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- (g) Find the Nash equilibrium of this game
 - (h) Illustrate the best response functions and the Nash equilibrium in a figure with q_1 on one axis and q_2 on the other axis.
 - (i) Find the total output (call it Q^*) that maximizes firms' total profit in the Cournot game. Compare $\frac{1}{2} Q^*$ with each firm's output in the Nash equilibrium, and show that firm's equilibrium profit is less than its profit in the "collusive" outcome, in which each firm produces $\frac{1}{2} Q^*$.
Why is the collusive outcome not a Nash equilibrium?

To maximize i 's profit which is

$$\max\{q_i (1 - q_i - q_{-i}) - c q_i\}$$

Taking the derivative and setting it equal to zero (FOC)

$$1 - 2q_i - q_{-i} - c = 0 \Leftrightarrow q_i(q_{-i}) = \frac{1 - c - q_{-i}}{2}$$

- (b) What is the most firm i would ever be willing to produce?

Because the smallest possible q_{-i} is zero, firm i would never produce more than

$$q_1(0) = \frac{1-c}{2}$$

(c) What strategies survive one round of elimination of strictly dominated strategies (IESDS)?

From b, it is clear that every quantity greater than $\frac{1-c}{2}$ is strictly dominated

Thus, only quantities in the interval $q_1 \in [0, \frac{1-c}{2}]$ survives one round of IESDS

(d) What strategies survive two rounds of IESDS?

Because we found that $q_1 \leq \frac{1-c}{2}$, firm 2 will also a quantity no less than $q_2 \leq \frac{1-c}{2}$

So, second round of IESDS implies

$$q_i\left(\frac{1-c}{2}\right) = \frac{1-c-q_2}{2} = \frac{1-c-\frac{1-c}{2}}{2} = \frac{1-c}{4}$$

Which means that

$$q_i \in \left[\frac{1-c}{4}, \frac{1-c}{2}\right]$$

(e) What strategies survive three rounds of IESDS?

Same principle as before: firm i will produce no less than a quantity of $\frac{1-c}{4}$ and therefore

$$q_i\left(\frac{1-c}{4}\right) = \frac{1-c-\frac{1-c}{4}}{2} = \frac{3-3c}{8}$$

(f) Determine the best response function of each firm.

Same as a)

$$q_i = BR_i(q_{-i}) = \frac{1-c-q_{-i}}{2}$$

(g) Find the Nash equilibrium of this game

Identical quantities in NE because of identical MC, so:

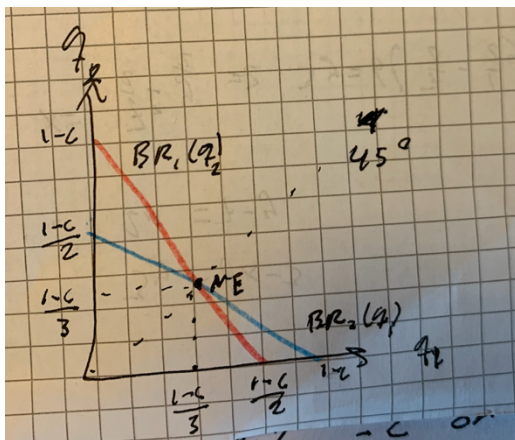
$$q_i = BR_i(q_{-i}) = \frac{1 - c - q_{-i}}{2} \Leftrightarrow q_i = \frac{1 - c}{3}$$

The NE is therefore

$$(q_1, q_2) = \left(\frac{1 - c}{3}, \frac{1 - c}{3} \right)$$

- (h) Illustrate the best response functions and the Nash equilibrium in a figure with q_1 on one axis and q_2 on the other axis.

Identical slopes with q_1 on x-axis and q_2 on y (one as a function of q_1 and the other as a function of q_2) because of identical MC



- (i) Find the total output (call it Q^*) that maximizes firms' total profit in the Cournot game. Compare $\frac{1}{2} Q^*$ with each firm's output in the Nash equilibrium, and show that firm's equilibrium profit is less than its profit in the "collusive" outcome, in which each firm produces $\frac{1}{2} Q^*$.

Why is the collusive outcome not a Nash equilibrium?

Profit function is

$$\max\{Q(1 - Q) - cQ\}$$

FOC is

$$1 - 2Q - c = 0 \Leftrightarrow Q = \frac{1 - c}{2}$$

And in the FOC before was

$$q_1 + q_2 = \frac{1 - c}{3} + \frac{1 - c}{3} = \frac{2 - 2c}{3} > \frac{1 - c}{2} = Q$$

Which holds for every $c < 1$

Which implies that competition makes the firms to produce more than the profit maximising quantity (the usual under monopoly, the quantity produced is less than under competition)

Calculating the profits

$$\begin{aligned}
 \pi_i^{NE} &= q_i(1 - q_i - q_{-i}) - cq_i \\
 &= q_i(1 - 2q_i - c) \\
 &= \left(\frac{1-c}{3}\right)\left(1 - \frac{2-2c}{3} - c\right) \\
 &= \left(\frac{1-c}{3}\right)^2 \\
 &= \frac{(1-c)^2}{9}
 \end{aligned}$$

And the Cournot profit where $q_i = \frac{1}{2}Q$

$$\begin{aligned}
 \pi_i^{Coll} &= q_i(1 - q_i - q_{-i}) - cq_i \\
 &= \frac{1}{2}Q\left(1 - 2 \cdot \frac{1}{2}Q\right) - cQ \\
 &= \frac{1}{2}Q(1 - Q - c) \\
 &= \frac{1}{2} \cdot \frac{1-c}{2}\left(1 - \frac{1-c}{2} - c\right) \\
 &= \frac{1-c}{4}\left(1 - 2 \cdot \frac{1}{2} \cdot \frac{1-c}{2} - c\right) \\
 &= \frac{1-c}{4}\left(1 - 2 \cdot \frac{1-c}{4} - c\right) \\
 &= \left(\frac{1-c}{4}\right)\left(\frac{1-c}{2}\right) \\
 &= \frac{(1-c)^2}{8}
 \end{aligned}$$

Implying that $\pi_i^{NE} < \pi_i^{Coll}$

So why not just collude instead?

Because there would be an incentive to deviate (classic cartel theory and also PD. $Q = \frac{1-c}{2}$ is the optimal Q found in part i):

$$\begin{aligned}
 BR\left(\frac{1}{2}Q\right) &= \frac{1-c-q_{-i}}{2} = \frac{1-c}{2} - \frac{1}{2} \cdot \frac{1}{2}Q = \frac{1-c}{2} - \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1-c}{2} \\
 &= \frac{1-c}{2} - \frac{1-c}{8} = \frac{3(1-c)}{8} > \frac{1-c}{4}
 \end{aligned}$$